

ZOOLOGY XI

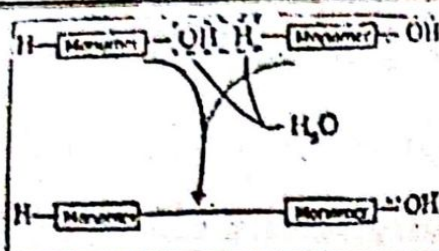
10 Year Questions Distribution

According to the Chapters.

This Distribution is carried out According to
NEW BOOK OF BIOLOGY.

COMPILED BY Dr. Abdul Raheem &
Miss. Sidra Mushtaq

QUESTIONS	YEARS
Chapter 1: BIOLOGICAL MOLECULES	
Section B	
2. (b) ix) Write different types of RNA. 2. (b) xvi) Explain Saturated and Unsaturated acylglycerol.	2025
2. (a)(ii) Why are water molecules called amphoteric in nature? 2. (a)(iv) Artificial sweetener is not harmful to diabetic patient. Why?	2024
2(a) iii) Why is artificial sweetener not harmful to diabetic patient? 2(a) iv) Why are water molecules called amphoteric in nature? 2(b) x) State the types of structural proteins with examples.	2023
2.(a) v) Why monosaccharides are sweet whereas polysaccharides are not?	2022
2. (b) x) Write note on condensation.	2021
2 (a) (iv) How are saturated acylglycerols different from unsaturated acylglycerols? 2 (b). (x) Describe the important functions of carbohydrates. 2 (b). (xvi) Identify the given process. Write its significance with examples.	2019



2. (b) (x) Differentiate between DNA and RNA	2018
(i) How is DNA different from RNA? (xvi) Write the names of different conjugated molecules with examples.	2017
(i) Write any four functions of Proteins. (vii) Write any two Biologically important properties of water. (xiv) Define Conjugated molecules & Explain any one of them.	2016
(xv) Write a note on structural types of Proteins.	2015
Section C	
5. What is polynucleotide? Define Watson and Crick model of DNA, draw structure of DNA model.	2025
5. —OR— Explain classification of protein and enlist the structural and functional protein.	
3. (OR) Define Lipids or Carbohydrates. Write their classifications and functions with examples.	2024
5. (b) Write characteristics and functions of Protein or Carbohydrate.	2022
3. Define Lipids. Write name of their different groups. Explain Terpenoids and their types.	2019
5. Define Carbohydrates. Describe their types and functions.	2018
3. Write the classification and functions of any one of the following: (i) Proteins (ii) Carbohydrates	2017

Chapter 2:

ENZYMES

Section B

2. (a) v) Why enzymes are called temperature sensitive?	2025
2. (b) xi) What are the inhibitors of enzymes? Explain with examples.	
2. (b) xv) Write a note on Lock and Key Model.	
2. (a)(iii) Why are enzymes temperature sensitive?	2024
2. (b)(xii) Write four characteristics of enzymes.	
2(b) viii) Write the characteristics of Enzymes.	2023
2.(b) ix) Write any four characteristics of enzymes.	2022

2. (a) (i) How does substrate concentration affect the activity of enzymes?	2021
2 (b). (xi) Write note on Lock and Key Theory.	2019
2. (a) (vii) How is the activity of enzyme affected by pH and temperature?	2018
2. (b) (ix) Write any four characters of Enzymes.	
(v) Name the factors which affect the activity of Enzymes.	2016
Section C	

Chapter 3: CELL STRUCTURE AND FUNCTION

Section B

2. (a) viii) Why do phospholipids form a bilayer in the cell membrane?	2025
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Chapter 7: PROTOCTISTS AND FUNGI

Section B

2 (b). (xii) Write the name of the class of the following protozoans: *Trypanosoma *Plasmodium *Entamoeba histolytica *Balantidium	2019
(viii) How are Protozoans differentiated by means of their locomotary organelles?	2017

Section C

3. Describe the characters of phylum Annelida OR Protozoa. Classify the phylum and give characteristics of any three classes with examples.	2015
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Chapter 9: DIVERSITY AMONG PLANTS

Section B

2. (a) i) Why kiwi cannot fly?	2025
2. (a) vii) Why diploblastic animals are different from the triploblastic?	
2. (b) x) Differentiate Polyp and Medusa	
2. (b) xiii) How reptiles are successful in land vertebrate?	
2. (b)(x) List the four basic characteristics of Chordates.	2024
2. (b)(xi) Differentiate between Polyp and Medusa.	
2. (b)(xiv) Write note on Subclass Ratitae and Carinatae.	
2. (b)(xv) Define Diploblastic and Triploblastic animals with examples.	

2(a) v) Why are Annelids, Mollusca and Arthropods considered as protostomes? 2(b) vii) Differentiate Cartilaginous and Bony fishes.	2023
2.(a) ii) Why is the blood of insects colorless? 2.(a) vi) Why kiwi cannot fly whereas owl can fly? 2.(b) viii) Write the names and functions of various types of canal system in porifera. 2.(b) x) Write note on polymorphism.	2022
2. (a) (ii) Why are amphibians successful land vertebrates? 2. (a) (vi) Why is Archaeopteryx considered as connecting link between Reptiles and Birds? 2. (b) viii) Differentiate between Chondrichthyes and Osteichthyes. 2. (b) xi) Write the parasitic adaptations in Helminthes.	2021
2 (a) (ii) Why do some animals undergo Ecdysis or moulting? 2 (a) (iii) Why is the blood of insects colorless? 2 (a) (vi) Why are Amphibians unsuccessful land vertebrates? 2 (b). (xiii) Write note on Lung fish. 2 (b). (xv) Write flight adaptation of birds.	2019
2. (a) (iii) Why can an eagle fly and penguin can not? 2. (a) (viii) Why are Amphibians unsuccessful vertebrates? 2. (b) (xii) Differentiate between Bony fish and Cartilaginous fish.	2018
(iii) How does Pearl formation take place in Oysters? (v) Why can Kiwi not fly? (ix) What are Parasitic adoptions in flatworms?	2017
(ii) What is Symmetry? (iii) Write any four characteristics of class Reptillia. (iv) What adaptation made the Birds capable of flight? (viii) Define Diploblastic and Triploblastic organization with examples. (x) Name the common edible Fishes of Pakistan. (xv) Define protostomata and Deutrostomata with examples.	2016
(ii) State any four distinguishing features of phylum Echinodermata. (x) Discuss the Economic importance of Arthropoda. (xi) Write four differences between Ratitae and Carinitae and give examples. (xii) Describe any four salient characters of phylum Porifera.	2015
Section C	
4. Describe the general characteristics of Class Mammals. Name its sub classes along with distinguishing characteristics	2025

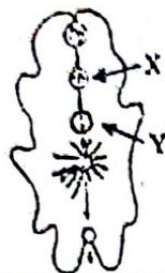
with example of each.	
4. Write the general characteristics, evolutionary adaptations and economic importance of Phylum Annelida OR Arthropoda.	2024
4. Write the general characteristics, evolutionary adaptation and economic importance of Phylum Mollusca or Arthropoda.	2023
3. (a) Write four salient features of Phylum Mollusca or Arthropoda. Classify it and write the characteristics of any three classes. 3. (b) Explain the characteristics of class Mammalia and write the names of sub-classes with examples.	2022
3. a) Describe the salient features of phylum Annelida OR mollusca. Classify the phylum and give the characteristics of each class. 3. b) Write the characteristics of any one class that belongs to Amniotes.	2021
3.[OR] Write the characteristics of Prototheria, Metatheria and Eutheria with examples. 5. Describe salient features of Phylum Annelida or Phylum Arthropoda. Classify it and write the characteristics of any three classes with examples.	2019
4. Describe the salient features of phylum Arthropoda OR Mollusca. Classify them and write the characters of any three classes with examples.	2018
4. Describe the salient features of phylum Annelid, classify the phylum and write the characteristics of each class with example. OR 4. Describe the characteristics of the most complex and successful group of vertebrates. Classify into sub classes and write the characteristics of each sub-class with examples	2017
4. Describe the salient features of Phylum Mollusca OR Arthropoda. Classify them and write the characteristics of any three classes with examples.	2016

Chapter 11: HOLOZOIC NUTRITION

Section B

2. (a)(i) How does Bile help in digestion of fats?	2024
2. (a)(viii) How is stomach protected from HCl?	
2(a) ii) How does stomach protect against HCl?	2023
2. (b) (xi) Which function is performed at X and Y in the given	2018

diagram.



(xii) Define Peristalsis and Anti-peristalsis movement.

2017

Section C

5. (a) Write any four Gastro-intestinal disorders of man.

2022

4. a) Describe the digestive system of man (Diagram is not required)

2021

4. Write the names of gastro intestinal tract disorders of man.

2019

3. Describe all the Cardiovascular disorders in man.

2018

5 [OR]. Describe the Digestive system of man with labelled diagram.

OR 5. List the names of Gastro-intestinal disorders in Man. Give description of any five of them.

2017

5. OR Define digestion. Explain the digestion of proteins and lipids in the digestive tract of Man.

2016

OR 4. List Cardiovascular Disorders. Write the causes, symptoms and precautionary measures of any five disorders mentioned in the list.

2015

5. Describe various disorders of Gastrointestinal tract in human being.

Chapter 12: CIRCULATION

Section B

2. (a) ii) Why circulatory system of Arthropods is called open type?

2025

2. (a) iv) Why blood flow faster in Arteries?

2. (b) xiv) Why electrocardiogram test performed?

2. (a)(vi) Why is S.A. Node called pacemaker of the heart?

2024

2. (b)(ix) Mention the role of arteriole in vasodilation and vasoconstriction.

2. (b)(xvi) Differentiate between single and double circuit circulation with examples.

2(a) i) Why is SA node called pace maker of the heart?

2023

2(a) vi) Why do capillaries have single layer of Endothelium?

2(b) ix) Write a note on ECG.

2.(a) i) Why do Leucocytes are known as soldiers of the body?	2022
2. (a) (iv) Why is a double circuit heart in animals better than a single circuit heart?	2021
2. (b) x) Write the functions of lymphatic system.	
2 (a) (viii) Why are valves absent in the lumen of artery?	2019
2. (b) (xiv) Differentiate between Artery and Vein.	2018
(xii) Write the functions of Human blood.	2016
(i) Write the functions of Lymphatics System.	2015

Section C

5. (OR) Describe Thrombus, Atherosclerosis and Angina Pectoris. Explain coronary bypass and Angioplasty.	2024
3. (OR) Describe the structure of Human Heart with the help of labelled diagram. Explain heart attack and heart failure.	2023
4. a) (OR) Explain Adaptive Immune System.	2021
5. a) Describe the internal structure of the human heart and draw its neat and labelled diagram.	
5. b) Differentiate between Artery and Vein.	
5. Describe the internal structure of the Human heart and mechanism of blood circulation through it with the help of diagram.	2017
4. Explain the structure of Human heart. Draw neat and labeled diagram of the internal structure of Human heart. OR	2015
4. List Cardiovascular Disorders. Write the causes, symptoms and precautionary measures of any five disorders mentioned in the list.	

Chapter 13: IMMUNITY

Section B

2. (a) iii) Why does each antibody bind only to a specific antigen?	2025
2. (a) vi) What are the causes of body's inflammatory response?	
2. (a) v) Why in Antipyretic therapy used?	2024
2(b) xii) Write a note on Allergy or Autoimmune disease.	2023
2 (b). (xiv) Differentiate between Active immunity and Passive immunity	2019
(xiv) Differentiate between Active and Passive immunity.	2015

Section C

3. ---OR---What are Autoimmune disorder? Explain the role of T-cells and B-cells in transplant rejections.	2025
5. Explain first line of defense with reference of Digestive Tracts and Respiratory Tracts.	2024
3. What is Cytotoxicity? Explain the mechanism of NK cells and TC cells.	2023
4. (a) Explain Innate Immune system.	2022
4. (b) Differentiate active immunity and passive immunity	
4.[OR] Define immune system. Explain Innate immune system.	2019
3. Define Immunity. Give an account of Innate OR adaptive immune system.	2016

Chapter 14: GASEOUS EXCHANGE

Section B

2. (b) xii) Differentiate Inspiration and Expiration:	2025
2. (a)(vii) Why is haemoglobin called as respiratory pigment?	2024
2. (b)(xiii) Define Sinuses and write their types.	
2(b) xi) Mention the causes of Sinusitis or Otitis Media.	2023
2.(a) iv) How does smoking cause lung cancer?	2022
2. (a) (v) Why is air better respiratory medium than water?	2021
2. (b) ix) Draw a neat and labelled diagram of Inspiration and Expiration in Man (ventral view).	
2. (b) ix) (OR)Write the conditions necessary for a respiratory surface in animals.	
2. (a) (v) How does Emphysema increase the residual volume of lung?	2018
(iv) How does smoking cause Lungs cancer in Man?	2017
(xi)What is the role of Hemoglobin and Myoglobin in the human body	
(xi) Write a note on Lung Cancer.	2016
(iv) State the causes & symptoms of Asthma OR Emphysema.	2015

Section C

3. Describe the breathing process in man. Explain the transport of Carbon dioxide and oxygen.	2025
3. Explain lower respiratory tract diseases.	2024
5. Explain main components of the respiratory system of man and draw neat and labelled diagram.	2023
5. Describe the Breathing mechanism in Human being. Explain the transport of CO ₂ and O ₂ .	2016

BOTANY XI

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Chapter 3: CELL STRUCTURE AND FUNCTION	
Section B	
iv) Why chloroplast considered as an energy converting organelle? v) Write a note on Nucleus OR Mitochondria. viii) Differentiate between any One of the following: *Rough Endoplasmic Reticulum and smooth Endoplasmic Reticulum.	2025
i) Why Lysosome is called suicidal sac? ii) Why are chloroplast and mitochondria considered as semi-autonomous organelles?	2024
2. xiii) Explain the movement mechanism of Cilia and Flagella. 2. xiv) Write significance of differential staining.	2023
2. (i) Mention any two differences between prokaryotic and eukaryotic cell. 2. (xi) Name two techniques used to isolate the component of cell. 3. (i) Draw a labelled diagram of fluid mosaic model of plasma membrane. 3. (ix) Write note on any one of the following: (a) Nucleus (b) Mitochondria	2018
2. (ii) State one function each of the following: (a) Ribosome (b) Mitochondria 2. (v) Name various components of the nucleus.	2017
ii) What is the cause of lysosomal storage disease? Name them. ix) Write a note on Nucleus or Endoplasmic reticulum.	2016
ii) Write a note on Mitochondria or plastids	2015

Section C

5. Explain the chemical composition, structure and functions of plasma membrane.	2025
4. (OR) Explain the structure and composition of Cell Wall.	2024
7. Explain the chemical composition of Plasma membrane. Also describe its functions in regulating cell's interaction with environment.	2023
4. (b) Describe Fluid Mosaic Model of plasma membrane. (Diagram not require)	2022
5. With the help of Fluid Moasic Model write the structure, properties and function of the plasma membrane.	2016

Chapter 4: BIOENERGETICS

Section B

x) Why ATP is known as common energy currency of living system?	2025
xi) What is the molecular formula of chlorophyll 'a' and 'b'?	
xiii) How C_4 Plants are different from C_3 plants?	
iii) Write a note on any one of the following: * CAM Plants * C_4 Plants	2024
iv) Why photorespiration considered as wasteful process?	
2. ix) Why ATP is called common energy currency of living system?	2023
2. (iv) What is ATP?	2022
2. (xi) Name first stable compound of Kreb cycle.	
3. (vi) Write a note on oxidative phosphorylation	
3. (viii) Describe role of water in light reaction.	
2. i) Why is ATP called Biological currency of energy?	2021
2. (ix) Why is ATP called the Currency of Energy?	2019
2. (x) Write the names of reactants and products of Photosynthesis.	
3. (viii) Draw a chart showing the efficiency of Food chain.	
3. (v) What is fermentation? Also name its types.	2018
2. (xv) Which compound receives CO_2 during dark reaction?	2017
2. (xvi) Name only Two types of fermentation.	
3. (iv) Why ATP called energy currency?	
xiv) Write a note on Oxidative phosphorylation.	2016
xv) Write the economic importance of fermentation.	
xvi) Write a note on Alternative mechanism of CO_2 fixation.	

iii) State the role of water OR chlorophyll in photosynthesis. x) State four important events take place during light dependent reaction of photosynthesis.	2015
Section C	
3. Explain light independent reaction of photosynthesis. 3. —OR— Explain oxidation of Pyruvic acid and kreb's cycle.	2025
4. Describe Cyclic and Non-cyclic phosphorylation during light reaction.	2024
4. Describe oxidative degradation of Pyruvic Acid. 4. (OR) Explain the light dependent reaction with reference to Electron transport chain and formation of ATP.	2023
5. (a) Describe the Glycolysis Pathway.	2022
6. Describe the light dependent reactions of photosynthesis.	2021
5. Define Cellular Respiration. Elucidate oxidative degradation of Pyruvic acid.	2019
6. Define Photosynthesis. Describe light reaction.	2018
7. With the help of a flow chart, describe the breakdown of glucose into pyruvate during glycolysis.	2017
5. OR Define a photosynthesis. Describe the light dependent reaction of photosynthesis	2016
5. Describe the process of Glycolysis. Also draw flowchart	2015

Chapter 5: ACCELLULAR LIFE

Section B

i) Write a note on Lytic cycle OR Lysogenic cycle. ix) Write a note on Viroids OR Prions. xiv) Write the stages of HIV life cycle.	2025
v) What are the symptoms of AIDS?	2024
2. viii) Classify Virus on the basis of Capsid.	2023
2. (v) Define Viroids.	2022
2. ii) Name any two diseases caused by Bacteria in human beings	2021
2. (ii) Name any two diseases caused by Virus. 3. (iv) Differentiate between Lysogenic and Lytic cycle. 3. (v) Write note on transmission of HIV virus. 2. (xvi) What are Prions?	2019
2. (iv) What is Bacteriophage? 3. (iv) Draw diagrammatic lytic cycle of bacteriophage	2018

2. (vi) Name any two diseases caused by virus.	2017
3. (iii) Draw a labeled diagram of Bacteriophage virus.	
vii) Write a note on transmission of HIV	2016

Chapter 6: PROKARYOTES

Section B

vi) Describe the classification of bacteria on the basis of shapes.	2025
vi) What are the various methods of control harmful bacteria?	2024
vii) Differentiate between Gram positive and Gram negative bacteria.	
2. (ii) Name types of bacteria on the basis of shape.	2022
3. (ii) Write salient features of Cyanobacteria.	
2. iv) Name three methods of genetic recombination found in Bacteria.	2021
3. iv) Write any four features of Cyanobacteria.	
3. vi) Classify Bacteria on the basis of flagella.	
2. (vii) Write only the name of the methods of Genetic recombination in Bacteria.	2019
2. (xiii) Draw a neat and labelled diagram of a Bacterial cell.	
3. (iii) Write the salient features of Cyanobacteria.	
2. (iv) Name the phases of Bacterial growth.	
3. (vii) Write four measures to control bacteria.	2018
2. (xv) What is Heterocyst ?	
2. (iii) How is bacterial cell wall different from plant cell wall?	2017
2. (vi) What is plasmid?	
3. (viii) How are bacteria classified on the basis of flagella?	
iii) Write the salient features of cyanobacteria.	2016
xiii) Methods of Genetic recombination in Bacteria.	
xiv) How bacteria is classified on the basis of Flagella.	2015

Section C

4. Describe the sources of nutrition in bacteria.	2025
6. What are Bacteria? Describe the structure of bacteria with the help of labelled diagram.	2024

Chapter 7: PROTOCTISTS AND FUNGI

Section B

viii) Differentiate between any One of the following: *Prokaryote and Eukaryote	2025
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viii) How prokaryotes are different from eukaryotes?	2024
ix) What features allow fungi to survive in all environments?	
2. iv) Name various divisions of fungi with one example each.	2023
2. (iii) Name various divisions of Fungi.	2022
2. (iii) Why is Deuteromycota called Fungi imperfect?	2019
3. (ix) Discuss the diversity of Kingdom Protoctista.	
2. (xiv) Name four groups of Fungi.	
2. (iii) Give two economic importance of Fungi.	2018
2. (x) What are lichens?	
2. (iv) Name various types of Ascocarp	2017
2. (vii) What are Mycorrhiza?	
3. (ii) Name only 4 groups of fungi with their reproductive structures.	
i) Name four groups of Fungi with their reproductive organs.	2016
xi) write note on lichen.	2015

Section C

6. (b) Describe and draw the life cycle of Zygomycota fungi	2022
5. Draw and describe the life cycle of Zygomycota.	2021
5. Draw and describe the life cycle of Ulva or Fern.	2018
6. Draw and describe the life cycle of Zygomycota or Ascomycota fungi.	2017
3. OR Describe the structure and reproduction in Ulva with reference to Isomorphic alternation of generation.	2015

Chapter 8: DIVERSITY AMONG PLANTS

Section B

x) Give botanical names of any two of the following: * Tulsi * Amaltas * Mulethi * Mako	2024
xi) Write three steps in evolution of seed.	
xii) Why cold treatment is necessary for germination of seed in some plants?	
xiii) Differentiate between Monocotyledon and Dicotyledon.	
xiv) Define Cymose Inflorescence.	
2. i) According to Webbed Limb, how many-veined leaves evolved?	2023
2. (viii) Name any three subdivisions of Tracheophyta.	2022
2. (ix) Name various classes of Bryophyta.	
2. (vii) Write botanical names of any two of the following:	

* Wheat * Rice * Red Pepper * Rose 3. (iii) Describe any one of the following: * Evolution of Seed * Evolution of Leaf 3. (vii) Draw any one of the following: * Marchantia Thallus	
2. (iii) Define Heterospory. 2. (vi) Name only the sub-divisions of Tracheophyta. 2. (vii) Why is Anthocerotae considered an advance class of Bryophyta? 3. (i) Draw well labelled diagram of any one of the following: * T.S of Marchantia Thallus 3. (ii) Why are Bryophytes called the amphibious plants. 3. (iii) Discuss the steps involved in evolution of seed.	2021
2. (v) Write the botanical names of any two of the following: * Rice * Potato * Almond 2. (viii) What is Double fertilization? 2. (xii) Write the names of five groups of Vascular plants. 3. (vii) Write note on any one of the following: * Evolution of Leaf * Evolution of Seed	2019
2. (vii) Write the botanical names of any two of the following: (a) Tobacco (b) Edible pea (c) Rose (d) Rice 2. (xiv) Mention three steps of evolution of seed in plants. 3. (ii) Trace the evolution of leaf in plants. 3. (iii) Give the outline of the classification of Kingdom Plantae.	2018
2. (viii) Define Heterogamy. 2. (xii) Write the botanical names of any two of the following: * Wheat * Apple * Tomato * Amaltas 2. (xiii) Name different parts of a Carpel. 3. (i) Name only the sub-division of Tracheophyta. 3. (ix) Draw a labeled diagram of any one of the following: (ii) T/S of Marchantia thallus	2017
iv) Write the botanical names of any four of the following: * Brinjal * Tomato * Pear * Sweet pea * Rice vi) Name major groups of vascular plants with examples. vii) Write a note on the Evolution of Leaf or Seed.	2016
ix) Write down the botanical name of any four of the following. a) Rice b) Potato c) Tomato d) Wheat e) Tamarind xiii) Draw the labeled diagram of any one of the following: * L.S Male Cone of Pinus * L.S of Ovule of an anghiospermic plants xv) Draw an outline of classification of kingdom of plantae	2015
Section C	

3. Describe the life cycle of fern OR Flowering plants.	2024
3. Draw and describe the life cycle of Moss or Fern.	2023
4. (a) Draw and describe life cycle of Moss.	2022
4. Describe the life cycle of fern. (Diagram no required)	2021
6. [OR] With the help of diagram, describe the life cycle of Moss.	2019
4. Describe the life cycle of Fern	2016

Chapter 10: FORMS AND FUNCTIONS IN PLANT

Section B

ii) Define Halophytes. Write two examples of halophyte plants. iii) Describe the role of Sulphur OR phosphorus in Plants. vii) Why carnivorous plants use insects as food? xii) Define any One of the following: *Annual ring *Turgor pressure *Parenchyma	2025
2. ii) Write the role of any one macro nutrient in plant growth. 2. iii) How plants cope with High temperature? 2. v) Enlist any four adaptations in xerophytes to cope with water shortage. 2. vi) Write a note on different pathways for water to enter into Xylem. 2. vii) Name structural components of xylem and phloem. x) How annual rings are formed? 2. xi) Name various groups of plants on the basis of osmotic adjustment. 2. xii) Mention the role of Apical meristem.	2023
2. (xii) Define Ascent of Sap. 3. (v) Define Facilitated Diffusion 3. (iv) Mention the deficiency symptoms of Nitrogen in Plants.	2022
2. ix) Define Imbibition OR Osmosis. 3. vii) Mention any two factors which affect the opening and closing of stomata.	2021
2. (i) Name the pathways available for water to enter the Xylem. 3. (i) State Root Pressure Theory. 3. (vi) Define any two of the following: *Plasmolysis *Diffusion *Water potential	2019
2. (viii) Give two symptoms of nitrogen deficiency in plants. 3. (viii) Why do insectivorous plants trap insects while they also prepare carbohydrates by Photosynthesis ? 2. (ix) Define facilitated diffusion.	2018

2. (ix) Name three pathways of water absorption in plants.	2017
2. (xiv) Define Ascent of Sap	
viii) Describe the deficiency symptoms of Nitrogen or Potassium in Plants.	2016
x) What are the three different pathway available in root for water to enter Xylem?	
xi) Define any two of the following: Plasmaolysis b) Osmosis d) Active transport	
xii) Define Transpiration. Describe its various types in Plants.	
v) Describe the importance and deficiency symptoms of nitrogen in plants.	2015
xii) Explain source – Sink Movement.	
xvi) define any two of the following *Plasmolysis*Diffusion *Water potential*Imbibitions	
Section C	
6. Describe source-sink movement. How pressure flow hypothesis explain the mechanism of translocation of food?	2025
5. Define Ascent of Sap. Describe the cohesion and tension theory of water movement through xylem.	2024
5. What are growth regulators? Give their names and explain the role of Auxins?	2023
6. How does E. Munch hypothesis explain the translocation of food in plants?	
7. Discuss Transpiration pull theory.	2021
4. Define Transpiration. Describe the mechanism of Ascent of sap.	2018
5. OR Describe the mechanism of Ascent of Sap.	2015

MISCELLANEOUS QUESTIONS

QUESTIONS	CH	YEAR
3. v) Distinguish between any one of the following:		2021
* Angiosperm and Gymnosperm	8	
* Alcoholic Fermentation & lactic acid fermentation	5	
v) Distinguish between any one type:		2016
a) Angiosperm and Gymnosperm	8	
b) Photosynthesis and Respiration	13	
vi) Distinguish any one of the following:		2015
Aerobic Respiration and Anaerobic Respiration	13	
Angiosperm and gymnosperm	8	